Restaurant Market Expansion Screener

Product Requirements Document (PRD)

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Version 1.0 | 20 October 2026 | Status: Proposed next-version specification

Document purpose: Define the requirements and acceptance criteria for the next version. Existing features must be verified before being described as complete.

1. Executive Summary

The Restaurant Market Expansion Screener is a Google Sheets-based decision-support dashboard designed to help analysts compare restaurant markets across countries using a consistent set of metrics. It builds on an exploratory analysis of the Zomato restaurant dataset and defines a repeatable next version with transparent metric definitions, filter behavior, data-quality checks, and an explainable shortlist framework.

This PRD distinguishes existing project work from proposed enhancements. Features described as requirements are targets for validation and implementation; they should not be presented as completed until tested against the acceptance criteria in this document.

2. Background and Problem Statement

The exploratory dashboard helps users review restaurant data, but country comparisons can become inconsistent when users rebuild pivot tables or apply different definitions. Stakeholders need a shared view of market size, rating quality, and price positioning, with clear explanations of how each metric is calculated.

The dataset used for the portfolio project contains 9,551 restaurant records and country-description data covering 15 countries. These are dataset-level reference figures; the final unique-restaurant count and the number of eligible rated records must be verified after applying the documented cleaning and deduplication rules.

3. Goals and Success Measures

Goals

- Provide a consistent side-by-side comparison of the 15 countries represented in the dataset.
- Reduce repetitive manual pivot-table rebuilding for common country-comparison questions.
- Make metric definitions, filtering behavior, and any shortlist ranking transparent and reproducible.
- Document data preparation and validation so another analyst can reproduce the results.

Success measures (baseline and target to be established)

- Time required for a stakeholder to answer a standard country-comparison question.
- Number of predefined business questions answerable from the dashboard without creating new pivots.
- Percentage of data-quality checks passed, with exceptions documented.
- Percentage of required dashboard tests passed during user acceptance testing (UAT).

4. Users and User Stories

- Strategy or product analyst: As an analyst, I want to compare countries by restaurant count, average rating, and price tier so I can screen markets consistently.
- Business stakeholder: As a stakeholder, I want to see a ranked shortlist with the reason for each score so I can understand the trade-offs behind a recommendation.
- Data owner: As a data owner, I want cleaning and metric rules documented so that the analysis is repeatable and auditable.

5. Scope and Boundaries

In scope

- Country-level scorecard with restaurant count, average rating among rated restaurants, and price-tier distribution.
- Country and price-tier filtering, with connected charts and pivot tables behaving consistently.
- Lookup-driven summary with a clear fallback for missing country keys.
- Methodology tab documenting cleaning, exclusions, metric definitions, and known limitations.
- A proposed, explainable country-screening score, subject to stakeholder validation of weights and assumptions.

Out of scope for this version

- Live or scheduled data refresh.
- User-level behavior, orders, revenue, profitability, or demand forecasting.
- Causal claims about market attractiveness or guaranteed expansion success.
- External market data such as population, competition, rent, or operating costs unless separately sourced and added.

6. Functional Requirements

FR-01 — Country scorecard	The dashboard shall show restaurant count and average rating for the selected country or comparison set. The rated-restaurant sample size should be visible wherever average rating is shown.
FR-02 — Price-tier distribution	The dashboard shall show the count or percentage of restaurants in each available price tier. The denominator used for each percentage must be explicit.
FR-03 — Filters and connected visuals	Country and price-tier slicers shall filter all connected charts and pivot tables that use the relevant source range. Formula-driven KPI cards and summary tables shall be tested separately; slicers must not be assumed to control formula outputs automatically.
FR-04 — Lookup fallback	The country summary shall use a lookup-driven approach such as XLOOKUP with IFNA (or an equivalent supported formula) so missing keys display a readable fallback instead of an error.
FR-05 — Methodology	A methodology tab shall list cleaning steps, reasons, affected fields, and any exclusions or assumptions.
FR-06 — Shortlist explanation (proposed enhancement)	If a ranked shortlist is implemented, the dashboard shall display the score, component metrics, scoring weights, and a plain-language reason for each rank. Weights must be labelled as proposed until validated by a stakeholder.
FR-07 — Data limitations	The dashboard shall disclose that restaurant listings and ratings alone do not measure demand, revenue, profitability, or expansion feasibility.

7. Metric Definitions

Restaurant count	Count of distinct restaurants after applying the documented deduplication rule. If the dataset has no reliable unique restaurant identifier, use a clearly documented proxy and disclose its limitation. Do not assume record count equals unique restaurant count.
Average rating	Arithmetic mean of the cleaned aggregate-rating field for records with a valid rating. Blank or explicitly unrated entries are excluded. Display the number of rated records used in the calculation.
Price-tier share	Number of eligible restaurants in a price tier divided by the number of eligible restaurants in the same country, multiplied by 100. If any tier is excluded or missing, document the denominator and exclusions.
Rating sample size	Count of records with a valid rating that contribute to the country-level average.
Optional screening score	A normalized weighted score combining market size, rating quality, and price-positioning fit. This is a decision aid, not a prediction of success; its weights and definition of price fit require stakeholder agreement.

8. Proposed Shortlist Method (Optional Next-Version Feature)

A transparent screening score may be added after the core dashboard and data checks pass. An illustrative starting point is: Market Score = 0.40 × Size Score + 0.35 × Rating Score + 0.25 × Price-Fit Score. These weights are examples only and must not be presented as validated business priorities.

- Normalize component metrics to a common scale before combining them.
- Define price-positioning fit with the stakeholder; do not assume that a higher or lower price tier is inherently better.
- Set a minimum rated-sample threshold or show a low-sample warning to avoid over-interpreting averages from small samples.

- Show component scores and sample sizes beside the overall score so users can challenge or interpret the ranking.

The current dataset can support descriptive comparisons of restaurant listings, ratings, and price tiers. It cannot, by itself, establish unmet demand, revenue potential, profitability, or the causal reasons a market would succeed.

9. Data Preparation and Quality Controls

- Preserve the raw data as a reference and perform transformations in a separate cleaned-data sheet.
- Trim leading and trailing spaces in text fields and standardize inconsistent labels where appropriate.
- Identify duplicate records using a documented key or proxy; review ambiguous matches before deleting them.
- Handle blank, invalid, and unrated values consistently. Keep unrated records distinct from valid zero ratings.
- Review missing country mappings, missing price tiers, and implausible cost values; flag issues rather than silently changing values.
- Document each cleaning step, its rationale, the affected columns, and any rows excluded from specific metrics.
- Reconcile counts and totals to the cleaned dataset before publishing the dashboard.

10. Acceptance Criteria and UAT Checklist

Dataset reconciliation	Dashboard counts reconcile with the cleaned dataset using the same deduplication rule; any difference is explained.
Country coverage	All 15 countries in the country-description/reference data are accounted for, including countries with zero eligible records if applicable.
Slicer behavior	Each supported country and price-tier filter combination updates every connected chart and pivot table correctly; formula-driven elements are tested independently.
Formula reliability	No unexpected #N/A, #REF!, or other formula errors appear in supported filter combinations. Missing keys show the documented fallback.
KPI correctness	Restaurant count, average rating, and rated sample size for selected countries match independent calculations from the cleaned data.
Spot checks	Manually validate at least five countries against filtered source records and independently calculated results.
Price-tier shares	Shares total approximately 100% when all eligible tiers are included; any gap is explained by missing or excluded values and rounding.
Methodology completeness	Cleaning steps, metric definitions, assumptions, exclusions, and known limitations are documented.
Shortlist transparency	If ranking is implemented, weights and component scores are visible, reproducible, and labelled as unvalidated until approved.

11. Risks, Assumptions, and Open Questions

- Uneven country sample sizes may make average ratings less reliable in smaller markets. Should a minimum rated-sample threshold be applied?
- Rating behavior may differ across countries, limiting direct comparability.
- Restaurant listings are not equivalent to active supply, customer demand, or market opportunity.
- The source dataset may contain duplicate, incomplete, or inconsistent records; counts depend on documented cleaning decisions.
- Which factors should matter most to stakeholders: market size, rating quality, price positioning, or additional external indicators?
- Who will approve the scoring weights and the definition of a suitable price-positioning fit?

12. Deliverables and Next Steps

- Google Sheets dashboard with documented country and price-tier filtering.
- Methodology tab with cleaning steps, definitions, exclusions, and limitations.
- UAT checklist and a record of test results.
- Optional country scorecard and ranked shortlist after the scoring model is agreed and tested.
- Short recommendation memo that distinguishes data-supported observations from assumptions and follow-up research.

Recommended sequence

– Confirm metric definitions and deduplication logic.

– Validate all dashboard charts, pivots, slicers, and formula-driven KPI elements.

– Complete UAT and document exceptions.

– Agree on scoring criteria and weights before publishing a ranked shortlist.

– Present the findings with sample-size caveats and clear limitations.

Appendix A. Portfolio Positioning

Suggested portfolio description: “Built a Google Sheets restaurant-market screening dashboard using a Zomato dataset. Cleaned and summarized restaurant records, compared country-level ratings and price tiers, and defined QA criteria for consistent filtering and transparent market screening.”

Only claim features as completed when they are implemented and verified. Label the scoring model, automation, or other planned capabilities as proposed enhancements until finished.